

Module 1: Assignment

Security concept relationships, security controls, and security design patterns

MCY 630 Security Architecture

Assignment Submission Instructions

Submission requirements: assignment

File format: MSWord or pdf (preferred)

Report requirements

- Use the provided fillable PDF file to complete the assignment.

Questions

Consider the following case scenario

A local auto insurance company MCY manages the insurance contracts for clients using a simple file server application. The file server application does not enforce authentication/authorization for user access, and allows the insurance agents to view, create, upload, modify, and delete contract files, given that they know the IP address of the server. The contracts are documents which are used by the insurance agents to provide insurance coverage services to the clients, such as, to create quotes, generate premiums, apply policy coverage, and process reimbursements. The file server application is installed on one of the desktop computers, which is physically located in the 'utility room' in the office, along with other common-use devices, such as the printer, Wi-Fi router, network hub, etc.

1. (20 points) Considering the relationship flow of security concepts below, provide two distinct examples of the relationship flow, starting with a probable threat agent and finishing with the suitable control.

Threat Agent: **A** -> triggers -> Threat: **B** -> exploits -> Vulnerability: **C** -> materializes -> Risk: **D** -> may damage -> Asset: **E** -> results in -> Exposure: **F** -> can be countered by -> Control: **G**

2. (15 points) Considering the two distinct control examples in your response to Q1, identify the type and function for each of the security controls presented in your response. Justify your response with a short explanation for each security control with respect to the type and functionality.
3. (15 points) Considering the two distinct threat/vulnerability examples in your response to Q1, identify the suitable security design pattern(s) which may avoid or reduce each of the risks presented in your response. Justify your response with a short explanation for each threat/vulnerability with respect to the chosen security design pattern(s).